$DB = 10\text{m}$ and $OC = \frac{1}{2}(5) = 2.5$ m

$CD = OC - 1 = (2.5\text{ m}) - 1\text{ m} = 1.5$ m

$$x_1 = \sqrt{(10)^2 + (1.5)^2} = \sqrt{100 + 2.25}$$

$$= \sqrt{102.25} = 10.1 m$$

In a right triangle EFB,

$DB = 10\text{m}$ and $OE = \frac{1}{2}(5) = 2.5\text{m} = FA$

$FB = FA + AB = (2.5\text{ m}) + 1\text{ m} = 3.5$ m

$$x_2 = \sqrt{(10)^2 + (3.5)^2} = \sqrt{100 + 12.25}$$

$$= \sqrt{112.25} = 10.6 m$$

The path difference $\Delta x = x_2 - x_1 = 10.6\text{ m} - 10.1\text{ m} = 0.5$ m. Required that this path difference

$$\Delta x = \frac{\lambda}{2} = 0.5 \Rightarrow \lambda = 1.0\text{ m}$$

To obtain the frequency of source, we use

$$v = \lambda f \Rightarrow f = \frac{v}{\lambda} = \frac{343}{1} = 343\text{ Hz}$$

$$= 0.3\ k\text{Hz}$$

Note If the speakers were connected such that already the path difference is $\frac{\lambda}{2}$. Now, the path difference combines with a path difference of $\frac{\lambda}{2}$. This gives a total path difference of λ which means, the waves are in phase and there is a maximum intensity at point B.

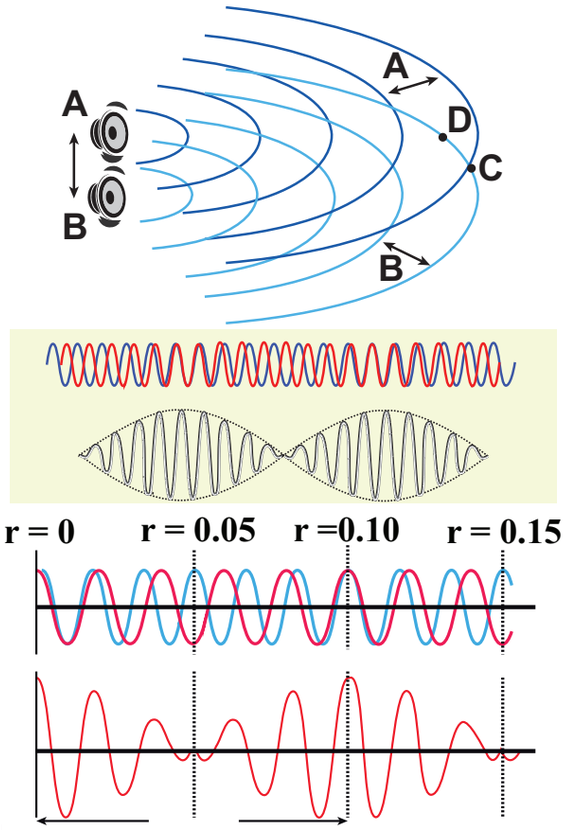


Figure 11.33: Two waves superimpose with different frequencies such that there is a time alternation in constructive and destructive interference i.e., they are periodically in and out of phase

11.7.2 Formation of beats

When two or more waves superimpose each other with slightly different frequencies, then a sound of periodically varying amplitude at a point is observed. This phenomenon is known as beats. The number of amplitude maxima per second is called beat frequency. If we have two sources, then their difference in frequency gives the beat frequency.

Number of beats per second

$$n = |f_1 - f_2| \text{ per second}$$

UNIT-11(XI-Physics_Vol-2).indd 252 26-03-2022 17:24:56